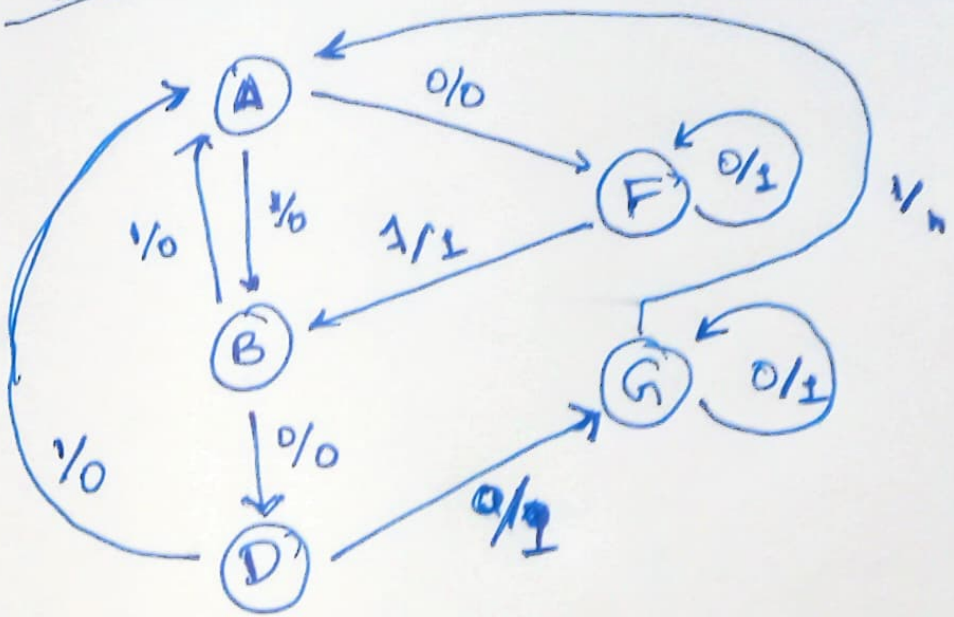


State transition table :-

Present state	Next $x=0$	Next $x=1$	Output $x=0$	Output $x=1$
S_0	S_3	S_2	0	0
S_1	S_2	S_0	0	0
S_2	S_4	S_0	1	0
S_3	S_3	S_1	1	1
S_4	S_4	S_2	0	1